NYT BESTSELLER PREDICTOR

Predicting New York Times bestseller potential based on publisher strength, author history, reviews, and timing.

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PROJECT OVERVIEW:

This project builds a machine learning model to predict a book's likelihood of appearing on the New York Times Best Seller list. Using features drawn from publisher reputation, author’s history, critical reviews, release timing, and retail demand signals. The model identifies patterns that distinguish bestsellers from comparable titles. The goal is to give publishers and authors a data-driven tool to evaluate a manuscript's commercial potential early in the publication pipeline.

MOTIVATION:

Book publishing is a $29.9B industry making high-stakes decisions on gut feel.

The publishing industry spends billions trying to answer this question, yet, it remains largely on guesswork.

Publishers, agents, and authors have no reliable tool to predict whether a book will reach the NYT list before it launches.

Combines NLP, classification ML, and multi-source data collection

TARGET AUDIENCE :

- Independent authors evaluating launch strategy
- Small publishers
- Literary agents (manuscript potential)
- Book marketers

DATA COLLECTION - API'S

NYT Books API
Bestseller list history since 2008, rank, weeks on list, category. This is the target variable.
Free API | CSV

Google Books API
Genre, publisher, page count, description text, publication date, author info
Free API | CSV

Open Library API
Author full bibliography, series vs. standalone, and community ratings.
Free API

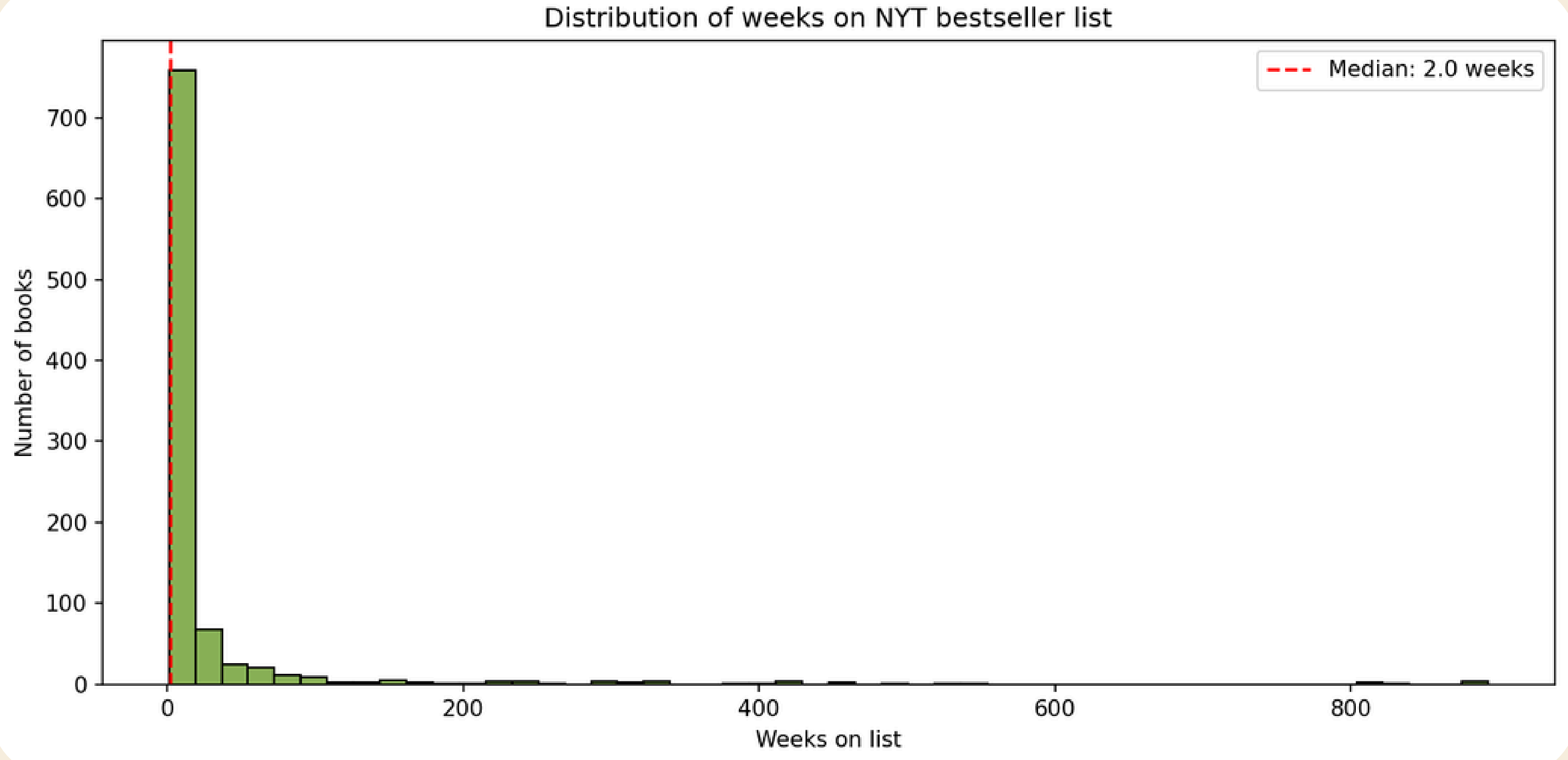
Spotify API
Title, author, and description
Free API

Total NYT rows:	6030
Unique books:	956
Matched Google Books records:	932/956
Date range:	26 weeks

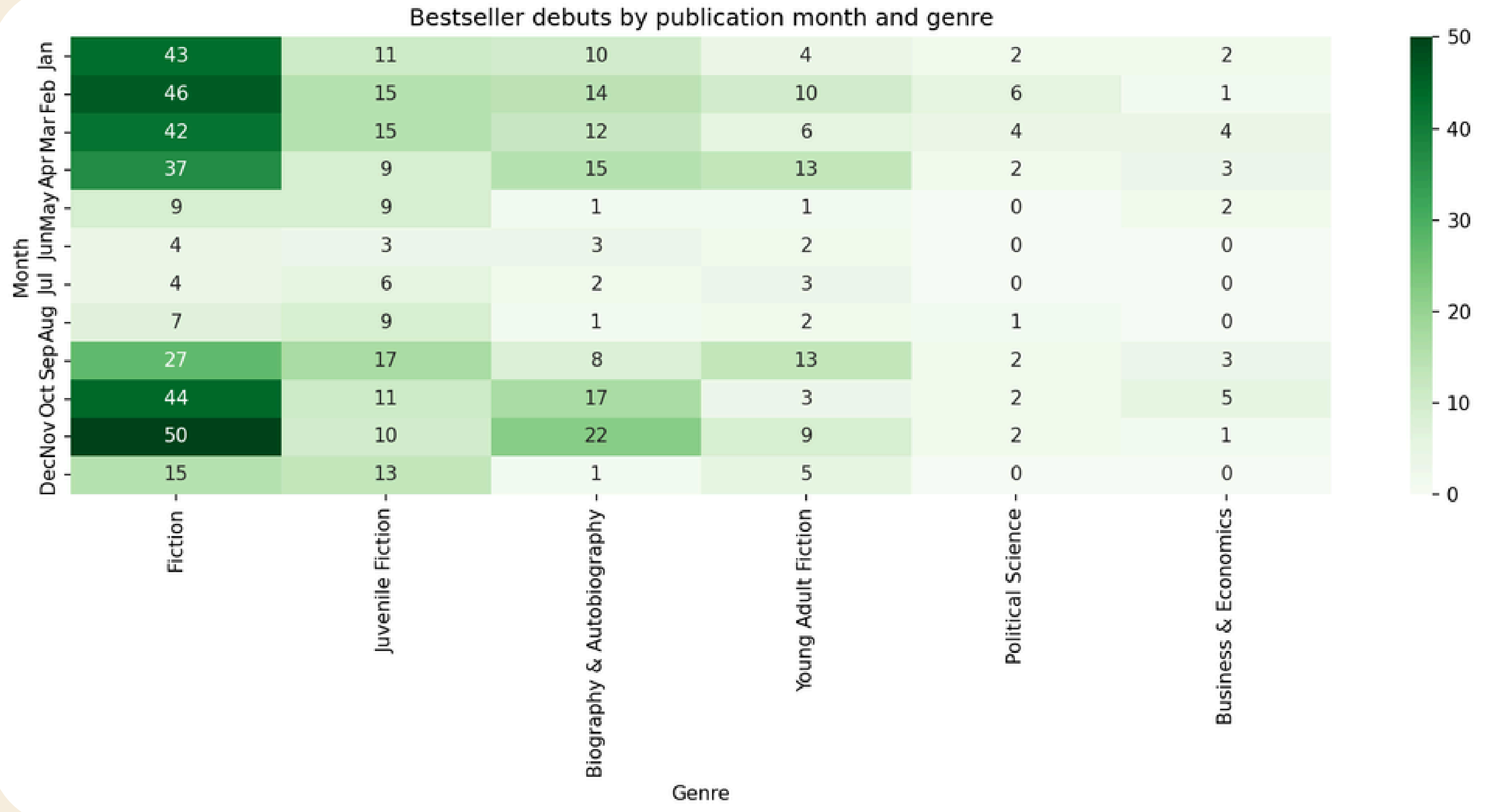
```
import requests, pandas as pd

class NYTBooksCollector:
    def __init__(self):
        self.api_key = os.getenv("NYT_API_KEY")
        self.base_url = "https://api.nytimes.com/svc/books/v3"
        # Rate limit safety settings
        self.min_request_interval = 12
        self.last_request_time = 0
```

EDA: NYT BESTSELLERS + GOOGLE BOOKS



Most NYT bestsellers are short-lived ~ 2 weeks



Market Timing is a real signal

Collected NYT bestseller data across multiple weeks and enriched book records with Google Books metadata.

Data quality: The book title vary across sources so I needed to use the ISBN number. Google Books categories were used for genre classification, so genre labels reflect Google's metadata rather than NYT list categories. Google Books matching was partial due to API rate limits.

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PROPOSED MODELING APPLICATION DESIGN

Target: Identify Potential NYT bestsellers

Secondary Target: The model can also estimate likely peak rank and total weeks on list

Logistic Regression

Good baseline classifier model. Will be used for interpretability and comparing the data.

XGBoost

Main model. This will help mixed feature types (numeric + categorical)

NLP Layer

VADER sentiment. This will help convert text into sentiment scores (positive, negative, neutral).

Features: Genre, Publisher tier, Author Prior NYT appearances, Publisher's Weekly Starred Reviews

Model Predictions: Bestseller Probability, Predicted Peak Rank, Predicted Weeks on List

Evaluation Metrics: Accuracy, Precision, Recall, F1 Score

Data Collection → Feature Engineering → Modeling → Evaluation / Prediction →

PROPOSED TIMELINE & MILESTONES

Week 6

Complete all Scraping + Merge Sources

Week 7

Full EDA + Feature Engineering

Week 8

Model training + Evaluation + Architecture Diagram

Week 9

Unit Tests + Deploy to Google Cloud Run + API live on public URL + Streamlit App Connected and Hosted

Week 10

GitHub Actions, README

Challenges & Mitigation:

- Title Mismatches → use ISBN
- Google Books API daily limit hit → cache all responses locally as JSON
- Class imbalance: far more non-bestsellers than bestsellers → use class_weight in XGBoost